Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2025

Course: B.Tech | *Time: 03 Hours*

Code: 100401 | *Subject: Biology for Engineers* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.

*(ii) There are **NINE** questions in this paper.*

*(iii) Attempt **FIVE** questions in all.*

(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** [2 x 7 = 14]

(a) Louis Pasteur discovered_____

(i) Light Microscope (iii) SEM

(ii) Pasteurization (iv) Sterilization

(b) Which of the following is not one of the steps acquired in the classification of living organisms?

(i) Recognizing the problem (iii) Formulating a hypothesis

(ii) Gathering data through experiments (iv) Publishing it for cross check

(c) Which of the following can be synonymous with growth?

(i) Meiosis (iii) Cell differentiation

(ii) Mitosis (iv) Cell Division

(d) Nucleoli are the subunits for_____

(i) Ribosome (iii) Mitochondria

(ii) Golgi apparatus (iv) Golgi Body

(e) What is the function of smooth ER (SER)?

(i) Protein production (iii) Provides smoothness to the cell

(ii) Provides attachment with mitochondria (iv) Synthesis and transport of lipids

(f) The simplest amino acid is

(i) Glycine (iii) Asparagine

(ii) Alanine (iv) Tyrosine

(g) The red colour of human blood is due to

(i) Myoglobin (iii) Immunoglobulin

(ii) Haemoglobin (iv) Haptoglobin

(h) Which of the following is not a component of a nucleotide?

(i) Pentose sugar (iii) Hexose sugar

(ii) Nitrogenous base (iv) Phosphate

(i) In which year the first cells were discovered?

(i) 1665 (iii) 1666

(ii) 1668 (iv) 1788

(j) Which of the following materials is commonly used in medical implants due to its biocompatibility and corrosion resistance?

(i) Steel (iii) Titanium

(iv) Aluminium

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store

Q.2	(a)	Write short notes on fundamental differences between science and engineering.	[7]
	(b)	Compare DNA and RNA.	[7]
Q.3	(a)	Compare Prokaryotic and Eukaryotic cell with suitable diagram.	[7]
	(b)	How autotrophs utilize carbon and energy.	[7]
Q.4	(a)	What are enzymes and explain its role in plants?	[7]
	(b)	What is Cellulose? Write any four functions of Cellulose.	[7]
Q.5	(a)	What is Mitotic Cell division? Explain Mitosis with neat diagram.	[7]
	(b)	Give brief account on Dominant Epistasis With suitable example.	[7]
Q.6	(a)	Compare Eye and Camera.	[7]
	(b)	Explain the Laws of Thermodynamics in biological systems.	[7]
Q.7	(a)	What is the concept of single celled organisms.	[7]
	(b)	Describe Glycolysis in detail with suitable diagram.	[7]
Q.8	(a)	Write a note on sterilization and various techniques used.	[7]
	(b)	Define Biology. Explain Terrestrial Habitat.	[7]

Q.9 Write short note on ***any two*** of the following: [7 x 2 = 14]

(a) Aquatic Habitat
(b) Photosynthesis
(c) Classification of microorganisms
(d) Growth Kinetics

BEU CONNECT
BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store